

Level 10 → Level 11

data.txt는 base64로 인코딩되어 있다고 한다. 그냥 열어보니 의미를 알 수가 없다. base64 -d 명령어로 디코딩하니 제대로 된 문장이 나왔다.

```
bandit10@bandit:~$ ls
data.txt
bandit10@bandit:~$ cat data.txt
VGhlIHBhc3N3b3JkIGlzIGR0UjE3M2ZaS2IwUlJzREZTR3NnMlJXbnBOVmozcVJyCg==
bandit10@bandit:~$ base64 -d data.txt
The password is dtR173fZKb0RRsDFSGsg2RWnpNVj3qRr
```

Level 11 → Level 12

data.txt는 알파벳을 13칸씩 미는 방식으로 암호화되어 있다고 한다. tr 명령어로 data.txt 안의 문자열을 치환하면 비밀번호를 얻을 수 있다.

```
bandit11@bandit:~$ ls
data.txt
bandit11@bandit:~$ cat data.txt
Gur cnffjbeq vf 7k16JArUVv5LxVuJfsSVdbbtaHGlw9D4
bandit11@bandit:~$ cat data.txt | tr 'A-Za-z' 'N-ZA-Mn-m'
The password is 7x16WNeHIi5YkIhWsfFIqoognUTyj9Q4
```

Level 12 → Level 13

문제에서 시키는대로 mktemp 명령어로 임시 디렉토리를 만들고 data.txt 파일을 복사했다.

```
bandit12@bandit:~$ mktemp -d
/tmp/tmp.iczzVq44bI
bandit12@bandit:~$ cd /tmp/tmp.iczzVq44bI
bandit12@bandit:/tmp/tmp.iczzVq44bI$ cp ~/data.txt .
```

data.txt 파일이 hexdump라고 하니 xxd -r 명령어를 이용해서 되돌려준다. 압축 형식으로 gzip, bzip2, tar가 나오니 알맞은 명령어를 이용해서 계속 압축을 풀었다.

```
bandit12@bandit:/tmp/tmp.iczzVq44bI$ xxd -r data.txt > data
bandit12@bandit:/tmp/tmp.iczzVq44bI$ file data
data: gzip compressed data, was "data2.bin", last modified: Fri Apr 3 15:17:36 2026, max compression, from Unix, original size modulo 2^32 576
bandit12@bandit:/tmp/tmp.iczzVq44bI$ mv data data.gz
bandit12@bandit:/tmp/tmp.iczzVq44bI$ ls
data.gz data.txt
bandit12@bandit:/tmp/tmp.iczzVq44bI$ gzip -d data.gz
bandit12@bandit:/tmp/tmp.iczzVq44bI$ file data
data: bzip2 compressed data, block size = 900k
bandit12@bandit:/tmp/tmp.iczzVq44bI$ mv data data.bz2
bandit12@bandit:/tmp/tmp.iczzVq44bI$ bzip2 -d data.bz2
bandit12@bandit:/tmp/tmp.iczzVq44bI$ file data
data: gzip compressed data, was "data4.bin", last modified: Fri Apr 3 15:17:36 2026, max compression, from Unix, original size modulo 2^32 20480
bandit12@bandit:/tmp/tmp.iczzVq44bI$ mv data data.gz
bandit12@bandit:/tmp/tmp.iczzVq44bI$ gzip -d data.gz
bandit12@bandit:/tmp/tmp.iczzVq44bI$ file data
data: POSIX tar archive (GNU)
bandit12@bandit:/tmp/tmp.iczzVq44bI$ mv data data.tar
bandit12@bandit:/tmp/tmp.iczzVq44bI$ tar xf data.tar
bandit12@bandit:/tmp/tmp.iczzVq44bI$ ls
data5.bin data.tar data.txt
bandit12@bandit:/tmp/tmp.iczzVq44bI$ file data5.bin
data5.bin: POSIX tar archive (GNU)
bandit12@bandit:/tmp/tmp.iczzVq44bI$ mv data5.bin data5.tar
bandit12@bandit:/tmp/tmp.iczzVq44bI$ tar xf data5.tar
bandit12@bandit:/tmp/tmp.iczzVq44bI$ ls
data5.tar data6.bin data.tar data.txt
bandit12@bandit:/tmp/tmp.iczzVq44bI$ file data6.bin
data6.bin: bzip2 compressed data, block size = 900k
bandit12@bandit:/tmp/tmp.iczzVq44bI$ mv data6.bin data6.bz2
bandit12@bandit:/tmp/tmp.iczzVq44bI$ bzip2 -d data6.bz2
bandit12@bandit:/tmp/tmp.iczzVq44bI$ file data6
data6: POSIX tar archive (GNU)
bandit12@bandit:/tmp/tmp.iczzVq44bI$ mv data6 data6.tar
bandit12@bandit:/tmp/tmp.iczzVq44bI$ tar xf data6.tar
```

ASCII 형식이 나올때까지 압축을 푼 후 파일을 읽어 비밀번호를 얻었다.

```
bandit12@bandit:/tmp/tmp.iczzVq44bI$ ls
data5.tar data6.tar data8.bin data.tar data.txt
bandit12@bandit:/tmp/tmp.iczzVq44bI$ file data8.bin
data8.bin: gzip compressed data, was "data9.bin", last modified: Fri Apr 3 15:17:36 2026, max compression, from Unix, original size modulo 2^32 49
bandit12@bandit:/tmp/tmp.iczzVq44bI$ mv data8.bin data8.gz
bandit12@bandit:/tmp/tmp.iczzVq44bI$ gzip -d data8.gz
bandit12@bandit:/tmp/tmp.iczzVq44bI$ file data8
data8: ASCII text
bandit12@bandit:/tmp/tmp.iczzVq44bI$ cat data8
The password is F05dwFsc0cbaIiH0h8J2eUks2vdTDwAn
```

Level 13 → Level 14

HINT 파일을 여니 localhost로 직접 접속은 막혔다고 한다. ssh.key.private을 여니 개인키가 나왔다.

```
bandit13@bandit:~$ ls
HINT sshkey.private
bandit13@bandit:~$ cat HINT
If you have trouble with this level, note the following:

1) As for all other levels, this level has a website with information:
   https://overthewire.org/wargames/bandit/bandit14.html
2) No, the level is not broken. To verify, see:
   https://status.overthewire.org/
3) The current version of OverTheWire prevents logging in from one
   level to another via localhost. Log out, and see 1)
4) If you get errors, read the error message on your screen.
   We mean it!
bandit13@bandit:~$ cat sshkey.private
-----BEGIN RSA PRIVATE KEY-----
MIIEpAIBAAKCAQEAXkkOE83W2cOT7IWhFc9aPaaQmQDdgzuXCv+ppZHa++buSkN+
gg0tcr7Fw8NLGa5+Uzec2rEg0WmeevB13AIoYp0MZyETq46t+jk9puNwZwIt9XgB
ZufGtZEwWbFWw/vVLNwOXBe4UWStGRWzgPpEeSv5Tb1VjLZIBdGphTIK22Amz6Zb
ThMsiMnyJafEwJ/T8PQO3myS91vUHEuo0MAzoUID4kN0MEZ3+XahyK0HJVq68KsV
ObefXG1vvA3GAJ29kxJaqrVfgYnqZryWN7w3CHjNU4c/2Jkp+n8L0Snxana+WYA7
jiPyTF0is8uzMlyQ4l1Lzh/8/MpvhCQF8r22dwIDAQABAoIBAQC6dWBjhyEOzjeA
J3j/RWmap9M5zfJ/wb2bfidNpwbB8rsJ4sZIDZQ7XuIh4LfygoAQSS+bBw3RXvzE
pvJt3SmU8hIDuLsCjL1VnBY5pY7Bju8g8aR/3FyjjNAqx/TLfzLLYf0u7i9Jet67
xAh0tONG/u8FB5I3LAI2Vp60viwvdWeC4nOxCthldpuPKNLA8rmMMVRTKQ+7T2VS
nXmwYckKUcUgzoVSpINZaS0zUDypdpY2+tRH3MQa5kqN1YKjvF8RC47wo0YCKtsD
o3FFpGNFec9Taa3Msy+DfQQhHKZFKIL3bJDONtmrVvtYK40/yeU4aZ/HA2DQzwe
o11AfiEhAoGBAOnVjosBkm7sblK+n4IEwPxs8s0mhPnTDUy5WGrpSCrXOmsVIBUf
laL3ZGLx3xCiwtCnEucB9DvN2HZkUpC/h6hTKUYLqXuyLD8njTrbRhLgbC9QrKrS
M1F2fStxVqPtZDlDMwjNR04xHA/fKh8bXXyTMqOHNJTHHNhbb3McdURjAoGBANKU
1hqfnw7+aXncJ9bjysr1ZWbqOE5Nd8AFgfwaKuGTTVX2NsUQnCMWdOp+wFak40JH
PKWkJNdBG+ex0H9JNQsTK3X5PBMAS8aFX0GrKeuwKWA6erytVTqj0fLYcdp5+z9s
8DtVCxDuVsM+i4X8UqIG0lvGbtKEVokHPFXP1q/dAoGAcHg5YX7WEehCgCYTzp0+
xysX8ScM2qS6xuZ3MqUWAXUWkh7NGZvhe0sGy9i0dANzwKw7mUUFVIAcMR/t54W1
GC83s0s3D7n5Mj8x3Nd08xFit7dT9a245Tva0YQ7KgmqpSg/ScKCw4c3eiLava+J
3btnJeSIU+8ZXq9XjPRpKwUCgYA7z6Li0QKxNeXH3qHXcnHok855maUj5fJNpPbY
iDkyZ8ySF8GLcFsky8Yw6fWCqfG3zDrohJ5l9JmEsBh7SadkwsZhvecQcS9t4vby
9/8X4jS0P8ibfckS4nBP+dT81kkkg5Z5MohXB0RA7VWx+ACohcDEkprsQ+w32xeD
qT1EvQKBgQDKm8ws2ByvSUVs9GjTilCajFqLJ0eVYzRPaY6f++Gv/UVfAPV4c+S0
kAWpXbv5tbkkzbS0eaLPTKgLvavXtQoTtKwrjpoLHKIHUz6Wu+n4abfAIRFubOdN
/+aLoRQ0yBDRbdXMsZN/jvY44eM+xRLdRVyMmdPtP8beLRi2E2aEzA==
-----END RSA PRIVATE KEY-----
```


exit로 bandit 서버에서 나온 뒤 nano 명령어로 위에 나온 RSA 개인 키를 붙여넣기 해서 sshkey.private 파일을 만든다. 소유자에게만 읽기, 쓰기 권한을 부여하는 chmod 600을 이용하고 이렇게 만든 키로 서버에 접속한다.

```
user@DESKTOP-CS2C1DH:~$ nano sshkey.private
user@DESKTOP-CS2C1DH:~$ chmod 600 sshkey.private
user@DESKTOP-CS2C1DH:~$ ssh -i sshkey.private bandit14@bandit.labs.overthewire.org -p 2220
```



This is an OverTheWire game server.
More information on <http://www.overthewire.org/wargames>

이제 비밀번호 파일을 읽을 수 있다.

```
bandit14@bandit:~$ cat /etc/bandit_pass/bandit14
MU4VWeTyJk8R0of1qqmcBPALh7lDCPvS
```

Level 14 → Level 15

현재 레벨(bandit14)의 비밀번호를 localhost의 30000번 포트에 전송하면 다음 비밀번호를 알려준다고 한다. 그대로 전송했더니 비밀번호를 얻을 수 있다.

```
bandit14@bandit:~$ echo MU4VWeTyJk8R0of1qqmcBPALh7lDCPvS | nc localhost 30000
Correct!
8xCjnmgoKbGLhHFAZlGE5Tmu4M2tKJQo
```